

CYBERSECURITY PORTFOLIO

HANDS-ON LAB · VIRTUALIZATION SECURITY

Hyper-V Security Hardening

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Portfolio Edition

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Original technical content and evidence preserved.
Other participant names removed for public portfolio use.

```

PS C:\Users\labadmin> ^C
PS C:\Users\labadmin> ^C
PS C:\Users\labadmin> Install-Module PSWindowsUpdate -Force

NuGet provider is required to continue
PowerShellGet requires NuGet provider version '2.8.5.201' or newer to interact with NuGet-based repositories. The NuGet
provider must be available in 'C:\Program Files\PackageManagement\ProviderAssemblies' or
'C:\Users\labadmin\AppData\Local\PackageManagement\ProviderAssemblies'. You can also install the NuGet provider by
running 'Install-PackageProvider -Name NuGet -MinimumVersion 2.8.5.201 -Force'. Do you want PowerShellGet to install
and import the NuGet provider now?
[Y] Yes [N] No [S] Suspend [?] Help (default is "Y"): y
PS C:\Users\labadmin> Import-Module PSWindowsUpdate
PS C:\Users\labadmin> Get-WindowsUpdate

ComputerName Status      KB          Size Title
-----
hv-pair01      ----- KB2267602    1GB Security Intelligence Update for Microsoft Defender Antivirus - KB2267602 (...

PS C:\Users\labadmin> ^C
PS C:\Users\labadmin> ^C
PS C:\Users\labadmin> Get-MpComputerStatus

AMEngineVersion      : 1.1.26070.7
AMProductVersion     : 4.18.26070.9
AMRunningMode        : Normal
AMServiceEnabled     : True
AMServiceVersion     : 4.18.26070.9
AntispywareEnabled   : True

```

```

PS C:\Users\labadmin> Get-NetFirewallProfile | Select Name,Enabled

```

```

Name      Enabled
----      -
Domain    True
Private   True
Public    True

```

```

PS C:\Users\labadmin> Get-LocalGroupMember "Hyper-V Administrators"
PS C:\Users\labadmin> Get-LocalGroup

```

```

Private    True
Public     True

```

```

PS C:\Users\labadmin> Get-LocalGroupMember "Hyper-V Administrators"
PS C:\Users\labadmin> Get-LocalGroup
Get-LocalGroup ^C
PS C:\Users\labadmin> Get-LocalGroupMember "Administrators"

```

```

ObjectClass Name              PrincipalSource
-----
User         hv-pair01\labadmin Local

```

```

PS C:\Users\labadmin>

```

```
PS C:\Users\labadmin> Get-LocalGroupMember "Administrators"

ObjectClass Name                PrincipalSource
-----
User          hv-pair01\labadmin Local

PS C:\Users\labadmin> Get-VM

Name      State   CPUUsage(%) MemoryAssigned(M) Uptime          Status           Version
-----
app-tier  Off     0           0                 00:00:00        Operating normally 10.0
db-tier   Off     0           0                 00:00:00        Operating normally 10.0
Ubuntu-test Running 0         2048          00:15:14.4100000 Operating normally 10.0
web-tier  Off     0           0                 00:00:00        Operating normally 10.0
```

```
Administration: Windows PowerShell
+ FullyQualifiedErrorId : ObjectNotFound,Microsoft.HyperV.PowerShell.Commands.GetVMFirmware

PS C:\Users\labadmin> Get-VMFirmware -VMName "app-tier"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
app-tier On          MicrosoftUEFICertificateAuthority IPv4          {File, Drive, Network}

PS C:\Users\labadmin> Get-VMFirmware -VMName "db-tier"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
db-tier On          MicrosoftWindows IPv4          {Drive, Network}

PS C:\Users\labadmin> Get-VMFirmware -VMName "db-Ubuntu-test"
Get-VMFirmware : hyper-V was unable to find a virtual machine with name "db-Ubuntu-test".
At line:1 char:1
+ Get-VMFirmware -VMName "db-Ubuntu-test"
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (:) [Get-VMFirmware], VirtualizationException
+ FullyQualifiedErrorId : ObjectNotFound,Microsoft.HyperV.PowerShell.Commands.GetVMFirmware

PS C:\Users\labadmin> Get-VMFirmware -VMName "Ubuntu-test"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
Ubuntu-test On          MicrosoftUEFICertificateAuthority IPv4          {File, Drive, Network, Drive}

PS C:\Users\labadmin> Get-VMFirmware -VMName "web-tier"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
web-tier Off          MicrosoftWindows IPv4          {File, Drive, Network}
```

```
PS C:\Users\labadmin> Get-VMFirmware -VMName "Ubuntu-test"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
Ubuntu-test On          MicrosoftUEFICertificateAuthority IPv4          {File, Drive, Network, Drive}

PS C:\Users\labadmin> Get-VMFirmware -VMName "web-tier"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
web-tier Off          MicrosoftWindows IPv4          {File, Drive, Network}

PS C:\Users\labadmin> Set-VMFirmware -VMName "web-tier" -EnableSecureBoot On
PS C:\Users\labadmin> Get-VMFirmware -VMName "web-tier"

VMName SecureBoot SecureBootTemplate PreferredNetworkBootProtocol BootOrder
-----
web-tier On          MicrosoftWindows IPv4          {File, Drive, Network}

PS C:\Users\labadmin>
```

```
web-tier On      MicrosoftWindows IPv4      {File, Drive, Network}
```

```
PS C:\Users\labadmin> Get-VMNetworkAdapter -VMName "web-tier"
```

Name	IsManagementOs	VMName	SwitchName	MacAddress	Status	IPAddresses
Network Adapter	False	web-tier	LabInternal	00155D1E0703		{}

```
PS C:\Users\labadmin> Get-VMNetworkAdapter -VMName "app-tier"
```

Name	IsManagementOs	VMName	SwitchName	MacAddress	Status	IPAddresses
Network Adapter	False	app-tier	LabInternal	00155D1E0701		{}

```
PS C:\Users\labadmin> Set-VMNetworkAdapter -VMName "app-tier" -RouterGuard On
PS C:\Users\labadmin> Set-VMNetworkAdapter -VMName "web-tier" -RouterGuard On
PS C:\Users\labadmin> Set-VMNetworkAdapter -VMName "Ubuntu-test" -RouterGuard On
PS C:\Users\labadmin> Set-VMNetworkAdapter -VMName "db-tier" -RouterGuard On
PS C:\Users\labadmin> █
```

```
db-tier Off
```

```
PS C:\Users\labadmin> Get-VMNetworkAdapter -All
```

Name	IsManagementOs	VMName	SwitchName	MacAddress	Status	IPAddresses
LabInternal	True		LabInternal	00155D1E0700	{Ok}	
Network Adapter	False	app-tier	LabInternal	00155D1E0701		{}
Network Adapter	False	db-tier	LabInternal	00155D1E0702		{}
Network Adapter	False	Ubuntu-test	LabInternal	00155D1E0704	{Ok}	{}
Network Adapter	False	web-tier	LabInternal	00155D1E0703		{}

```
PS C:\Users\labadmin> Get-VMNetworkAdapter -All
```

Name	IsManagementOs	VMName	SwitchName	MacAddress	Status	IPAddresses
LabInternal	True		LabInternal	00155D1E0700	{Ok}	
Network Adapter	False	app-tier	LabInternal	00155D1E0701		{}
Network Adapter	False	db-tier	LabInternal	00155D1E0702		{}
Network Adapter	False	Ubuntu-test	LabInternal	00155D1E0704	{Ok}	{}
Network Adapter	False	web-tier	LabInternal	00155D1E0703		{}

```
PS C:\Users\labadmin> █
```